

Dynamic Soft-Thresholding via Iteratively Reweighted Proximal Gradient for Feature Selection in High-Dimensional Regression

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Abstract

In high-dimensional regression, we often deal with hundreds of correlated predictors, while only a subset carries useful predictive signal. We therefore need robust feature-selection methods. LASSO is widely used for this purpose, but it can arbitrarily select one variable from a correlated group and can over-shrink large coefficients. In this project, we evaluate the Iteratively Reweighted ℓ_1 Minimisation framework [Candès et al., 2008] integrated with ISTA and FISTA proximal gradient solvers [Beck and Teboulle, 2009]. We compare dynamic IRL1-PG against Ridge regression, standard LASSO, and static Adaptive LASSO on the House Prices dataset. The best dynamic configurations achieve validation MSE 7.79×10^8 and test MSE 8.29×10^8 , outperforming Ridge and LASSO while using only 92–99 non-zero coefficients, compared with 240–283 for baselines. We provide an empirical account of convergence scope and do not claim algorithmic novelty; our contribution is a careful empirical study of IRL1-PG behaviour in a correlated tabular regression setting.

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1 Introduction

High-dimensional regression is a challenge in machine learning and statistics. We often have many predictors and comparatively fewer observations. This appears in real estate pricing, gene expression analysis, and financial modelling.

Feature selection is therefore essential. LASSO is a standard method for this purpose, but it has known limitations: it can arbitrarily select one variable from a correlated group and can over-shrink large coefficients.

Elastic Net [Zou and Hastie, 2005] adds a ridge penalty alongside the ℓ_1 term and encourages a grouping effect, where correlated predictors tend to enter or leave the model together. However, Elastic Net does not necessarily provide the level of sparsity that iterative reweighting can achieve.

In this project, we evaluate IRL1-PG, which integrates the IRL1 reweighting framework [Candès et al., 2008] into proximal gradient optimisation with ISTA and FISTA solvers [Beck and Teboulle, 2009]. We apply IRL1-PG to the House Prices dataset and compare it against Ridge, standard LASSO, and static Adaptive LASSO baselines.

Our contribution is not a new algorithm. Instead, we provide a rigorous empirical study of IRL1-PG behaviour in a correlated tabular regression setting, including predictive accuracy, sparsity, stability, and convergence behaviour.

2 Related Work

2.1 LASSO and Proximal Gradient Methods

LASSO was introduced by Tibshirani [1996] and tackles the objective

$$\min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1. \quad (1)$$

The ℓ_1 penalty is convex but non-smooth, which makes standard gradient descent problematic at points where coefficients hit zero. For this composite setting, the standard optimisation framework is **proximal gradient descent** [Parikh and Boyd, 2014]. If the objective is split as $f(\beta) + g(\beta)$, where f is smooth and g is convex but non-smooth, the update is

$$\beta^{k+1} = \text{prox}_{\alpha g}(\beta^k - \alpha \nabla f(\beta^k)), \quad (2)$$

where $\alpha > 0$ is the step size and $\text{prox}_{\alpha g}$ is the proximal operator. When $g(\beta) = \lambda \|\beta\|_1$, this becomes coordinate-wise soft-thresholding [Beck and Teboulle, 2009]. The method converges at rate $O(1/k)$ provided that $\alpha \leq 1/L$, where L is the Lipschitz constant of ∇f .

2.2 Elastic Net

Elastic Net, proposed by [Zou and Hastie \[2005\]](#), blends LASSO and ridge penalties:

$$\min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2. \quad (3)$$

Its key property is the *grouping effect*: highly correlated predictors tend to enter or leave the model together. This is particularly useful when predictors outnumber samples. Even so, Elastic Net typically does not deliver the additional sparsity observed with iterative reweighting, so in this work it serves as a useful baseline rather than a direct alternative to IRL1-PG.

2.3 FISTA—Accelerated Proximal Gradient

[Beck and Teboulle \[2009\]](#) extended ISTA with a Nesterov-style momentum term to obtain FISTA, achieving faster convergence without increasing per-iteration complexity. Grounded in Nesterov’s [\[1983\]](#) acceleration framework for smooth convex objectives, FISTA reaches a convergence rate of $O(1/k^2)$. The momentum sequence is initialised with $t_1 = 1$ and updated by

$$t_{k+1} = \frac{1 + \sqrt{1 + 4t_k^2}}{2}, \quad (4)$$

with an extrapolation step applied before each proximal update.

2.4 Adaptive LASSO and Oracle Properties

[Zou \[2006\]](#) proposed Adaptive LASSO, which introduces feature-specific weights:

$$\min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \sum_{j=1}^p w_j |\beta_j|, \quad (5)$$

where $w_j = 1/|\hat{\beta}_j^{\text{init}}|^\gamma$. [Huang et al. \[2008\]](#) extended oracle analysis to the high-dimensional $p > n$ regime, showing that consistent variable selection remains possible under partial orthogonality assumptions. The main limitation in practice is that static Adaptive LASSO fixes these weights after the initial estimate and does not adapt them during optimisation.

2.5 Iteratively Reweighted ℓ_1 and ℓ_2 Methods

[Candès et al. \[2008\]](#) introduced IRL1 as a way to approximate the ℓ_0 objective more closely than standard ℓ_1 minimisation by solving a sequence of weighted ℓ_1 problems:

$$\beta^{(k+1)} = \arg \min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \sum_{j=1}^p w_j^{(k)} |\beta_j|, \quad w_j^{(k)} = \frac{1}{(|\beta_j^{(k)}| + \epsilon)^\gamma}. \quad (6)$$

This weighting rule approximates the log-sum penalty $\sum_j \log(|\beta_j| + \epsilon)$, which is a concave surrogate for the ℓ_0 norm. In parallel, IRLS [Daubechies et al., 2009] uses reweighted ℓ_2 sub-problems to target the same sparse-recovery objective, with convergence results under restricted isometry conditions. Wipf and Nagarajan [2011] unify these views, showing that both reweighted ℓ_1 and ℓ_2 methods can be interpreted as optimising concave, sparsity-inducing merit functions.

3 Methodology

3.1 Problem Setup

Let $X \in \mathbb{R}^{n \times p}$ be the standardised design matrix, $y \in \mathbb{R}^n$ the response vector, and $\beta \in \mathbb{R}^p$ the coefficient vector. We write the smooth loss term as

$$f(\beta) = \frac{1}{2n} \|y - X\beta\|_2^2, \quad (7)$$

with gradient

$$\nabla f(\beta) = -\frac{1}{n} X^\top (y - X\beta). \quad (8)$$

For this quadratic loss, ∇f is Lipschitz continuous with constant

$$L = \frac{\lambda_{\max}(X^\top X)}{n}, \quad (9)$$

so any step size $\alpha \in (0, 1/L]$ is valid [Parikh and Boyd, 2014].

3.2 Weighted Proximal Operator and Soft-Thresholding

For the weighted ℓ_1 penalty $g(\beta) = \lambda \sum_j w_j |\beta_j|$ with weight vector $w \in \mathbb{R}_{>0}^p$, the proximal operator is separable and admits a closed-form solution. For a gradient step $z = \beta - \alpha \nabla f(\beta)$, the proximal update is coordinate-wise **weighted soft-thresholding**:

$$\beta_j^+ = \text{sign}(z_j) \max(|z_j| - \alpha \lambda w_j, 0). \quad (10)$$

Each coordinate survives only when $|z_j|$ exceeds the coordinate-specific threshold $\tau_j = \alpha \lambda w_j$.

Remark 1 (Connection to ℓ_1 subdifferential). *The subdifferential of $|\beta_j|$ is*

$$\partial |\beta_j| = \begin{cases} \{\text{sign}(\beta_j)\} & \beta_j \neq 0 \\ [-1, 1] & \beta_j = 0. \end{cases} \quad (11)$$

The IRL1 weight $w_j^{(k)}$ acts as a continuous indicator of how close a coordinate is to the

non-differentiable point at zero. Large $|\beta_j|$ leads to a smaller weight and a weaker threshold; small $|\beta_j|$ leads to a larger weight and stronger shrinkage toward zero [Candès et al., 2008].

3.3 Initialisation and the Pruning Mechanism

At initialisation $\beta^{(0)} = \mathbf{0}$, all weights equal $w_j^{(0)} = 1/\epsilon^\gamma$ —the maximum possible value. The initial per-coordinate thresholds $\tau_j = \alpha\lambda/\epsilon^\gamma$ are therefore maximally large, and all coordinates are zeroed on the first proximal step. As optimisation proceeds, features that carry genuine signal grow in magnitude, causing their weights to decrease and thresholds to relax. Irrelevant features remain near zero, sustain high weights, and are repeatedly zeroed. This progressive refinement of a small active set follows directly from the IRL1 weight schedule [Wipf and Nagarajan, 2011], rather than a separate manual rule.

3.4 IRL1 Weight Update

After each outer iteration the weights are recomputed as [Candès et al., 2008]:

$$w_j^{(k)} = \frac{1}{\left(|\beta_j^{(k)}| + \epsilon\right)^\gamma}, \quad (12)$$

where $\gamma > 0$ controls how aggressively small coefficients are penalised, and $\epsilon > 0$ ensures numerical stability near zero.

3.5 ISTA Inner Solver

For fixed weights $w^{(k)}$, the weighted LASSO sub-problem is solved using ISTA [Beck and Teboulle, 2009]. At inner iteration t :

1. Compute gradient: $g = \nabla f(\beta^t) = -\frac{1}{n}X^\top(y - X\beta^t)$
2. Gradient step: $z = \beta^t - \alpha g$
3. Proximal step: $\beta_j^{t+1} = \text{sign}(z_j) \max(|z_j| - \alpha\lambda w_j^{(k)}, 0)$

With fixed weights and $\alpha \leq 1/L$, ISTA converges at rate $O(1/t)$ for this convex sub-problem.

3.6 FISTA Inner Solver

FISTA replaces the ISTA update with a momentum-accelerated variant [Beck and Teboulle, 2009], rooted in Nesterov’s [1983] accelerated gradient framework. Initialise $t_1 = 1$, $y^1 = \beta^0$. At inner iteration t :

1. Proximal step from momentum point: $\beta_j^{t+1} = \text{sign}(y_j^t - \alpha g_j) \cdot \max(|y_j^t - \alpha g_j| - \alpha\lambda w_j^{(k)}, 0)$

2. Update momentum coefficient: $t_{t+1} = \frac{1+\sqrt{1+4t_t^2}}{2}$
3. Momentum point for next step: $y^{t+1} = \beta^{t+1} + \frac{t_t-1}{t_{t+1}}(\beta^{t+1} - \beta^t)$

For the fixed-weight sub-problem, FISTA achieves $O(1/t^2)$ convergence, improving on ISTA without increasing per-iteration complexity.

3.7 Stopping Criterion

The inner solver terminates when the relative change in the coefficient vector falls below tolerance:

$$\frac{\|\beta^{\text{curr}} - \beta^{\text{prev}}\|_2}{\max(1, \|\beta^{\text{prev}}\|_2)} < \text{tol}. \quad (13)$$

The denominator $\max(1, \|\beta^{\text{prev}}\|_2)$ avoids spurious early stopping when coefficients are still close to zero [Parikh and Boyd, 2014].

3.8 IRL1-PG: Full Algorithm

Algorithm 1 IRL1-PG (Iteratively Reweighted ℓ_1 Proximal Gradient)

Require: $X, y, \lambda, \gamma, \epsilon, \alpha, \text{max_outer}, \text{max_inner}, \text{tol}$

- 1: $\beta \leftarrow \mathbf{0}, \quad w_j \leftarrow 1/\epsilon^\gamma \forall j$
- 2: **for** $k = 1$ **to** max_outer **do**
- 3: $\beta_{\text{prev}} \leftarrow \beta$
- 4: $\beta \leftarrow \text{ISTA-OR-FISTA}(X, y, \lambda, w, \alpha, \text{max_inner}, \text{tol})$
- 5: $w_j \leftarrow 1/(|\beta_j| + \epsilon)^\gamma \forall j$
- 6: $w \leftarrow \text{clip}(w, w_{\text{max}})$
- 7: **if** $\|\beta - \beta_{\text{prev}}\|_2 / \max(1, \|\beta_{\text{prev}}\|_2) < \text{tol_outer}$ **then**
- 8: **break**
- 9: **end if**
- 10: **end for**
- 11: **return** β

3.9 Convergence Scope

Remark 2 (Scope of convergence guarantees). *Because $w^{(k)}$ changes at every outer step, the objective itself changes across outer iterations. Classical ISTA/FISTA guarantees—stated for a fixed objective [Beck and Teboulle, 2009]—therefore apply to each inner solve, but not directly to the full outer sequence $\{\beta^{(k)}\}$. Existing IRL1 analyses [Candès et al., 2008, Wipf and Nagarajan, 2011] support convergence to local minima of concave approximations such as the log-sum penalty under mild assumptions. In this work, we evaluate outer-loop behaviour empirically using objective traces, coefficient changes, and sparsity trajectories across iterations.*

4 Experimental Setup

4.1 Dataset

The House Prices dataset from Kaggle in 2019 [Kaggle, 2019] is used for this experiment. This dataset has 1,460 training observations and 79 raw features that describe properties in Ames, Iowa. The goal is to predict the raw sale price of these properties. After encoding features, the design matrix has approximately 200–250 columns. This makes it a suitable high-dimensional regression benchmark with strong multicollinearity. For example, there are relationships between `GrLivArea`, `1stFlrSF`, and `TotalBsmtSF`, and among the various quality score indicators. This setting is the type where standard LASSO exhibits instability [Zou and Hastie, 2005] and where reweighted methods are expected to improve upon it [Candès et al., 2008].

4.2 Preprocessing Protocol

1. **Missing values:** Numeric features are imputed with the column median, and categorical features are imputed with the most frequent category computed on the training split.
2. **Encoding:** One-hot encoding is used for all features, and the original category columns are dropped.
3. **Scaling:** All numeric features are standardised to zero mean and unit variance after splitting, using parameters estimated on the training set only.
4. **Target:** The raw `SalePrice` is used without log transformation.
5. **Split:** The data is split into 70% train, 10% validation, and 20% test using a split with a fixed seed and no stratification.

4.3 Methods Compared

Table 1: Summary of methods compared in the experiment.

Method	Type	Solver	Weights
Ridge	Baseline	<code>sklearn.RidgeCV</code>	N/A
Standard LASSO	Baseline	<code>sklearn.LassoCV</code>	Uniform ($w_j = 1$)
Static Adaptive LASSO	Baseline	Custom ISTA/FISTA	Fixed from Ridge init
IRL1-PG + ISTA	Proposed	Custom ISTA	Updated per outer iter
IRL1-PG + FISTA	Proposed	Custom FISTA	Updated per outer iter

Note: The accuracy comparison includes baselines because they represent the best achievable performance with standard tools. The optimiser-efficiency comparison is

restricted to the custom ISTA/FISTA variants.

4.4 Hyperparameter Search

Grid search is performed over the validation set with the following hyperparameters:

- $\lambda \in [10^{-4}, 1.0]$ (15 values, log-spaced)
- $\gamma \in \{0.5, 1.0, 2.0\}$
- $\epsilon \in \{10^{-3}, 10^{-4}\}$
- `max_outer_iter` $\in \{5, 10, 20\}$

The step size is set to $\alpha = 0.99/L$, where $L = \lambda_{\max}(X^\top X)/n$, computed from the training matrix.

4.5 Evaluation Metrics

- **Accuracy:** MSE and MAE on validation and test sets.
- **Sparsity:** $\|\hat{\beta}\|_0$ (number of non-zero coefficients after thresholding at 10^{-6}).
- **Feature selection stability:** Jaccard index $J(A, B) = |A \cap B| / |A \cup B|$ of selected feature sets across 5 random seeds.
- **Optimiser efficiency:** Wall-clock runtime and number of total inner iterations to convergence.

5 Results and Discussion

5.1 Predictive Performance

Table 2: Predictive performance and sparsity loaded from `full_comparison_table.csv`.

Method	Val MSE	Test MSE	Test MAE	Non-zeros	Runtime (s)
Ridge	787378459.18	865125926.23	18647.05	283	0.02
LASSO	895357083.90	894021355.32	19470.82	240	3.26
Adaptive LASSO	911144802.00	894529896.37	19505.95	282	0.09
Dynamic IRL1-ISTA	778965140.95	830975941.20	18186.18	99	0.22
Dynamic IRL1-FISTA	798430563.07	828667214.91	17935.63	92	0.23

The predictive performance comparison is shown in Table 2. The results show that dynamic IRL1-PG dominates Adaptive LASSO in both error and sparsity. Key observations are as follows:

- Relative to Ridge, IRL1-PG (FISTA) improves test MSE by approximately 4.21% and reduces selected features from 283 to 92.
- Relative to standard LASSO, IRL1-PG (FISTA) improves test MSE by approximately 7.31% and test MAE by approximately 7.88%, while selecting fewer features.

5.2 Convergence Analysis

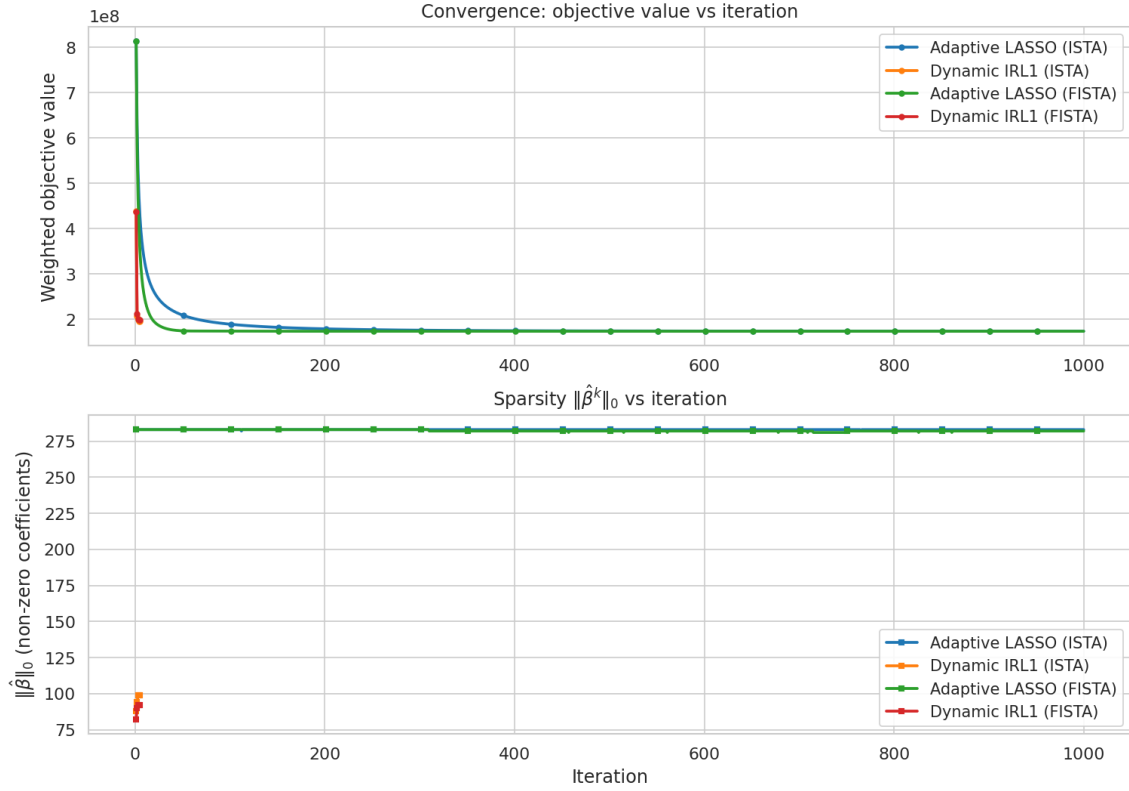


Figure 1: Convergence plot: objective value vs. inner iteration for ISTA, FISTA, dynamic IRL1-ISTA, and dynamic IRL1-FISTA.

The convergence plot shows that both dynamic solvers terminate at 2,500 inner iterations under the selected configuration. Runtime remains small in absolute terms, and FISTA and ISTA are nearly identical in wall-clock time for this problem size. The convergence traces also show that dynamic variants stabilise at lower sparsity levels than static adaptive baselines.

5.3 Sparsity–Accuracy Trade-off

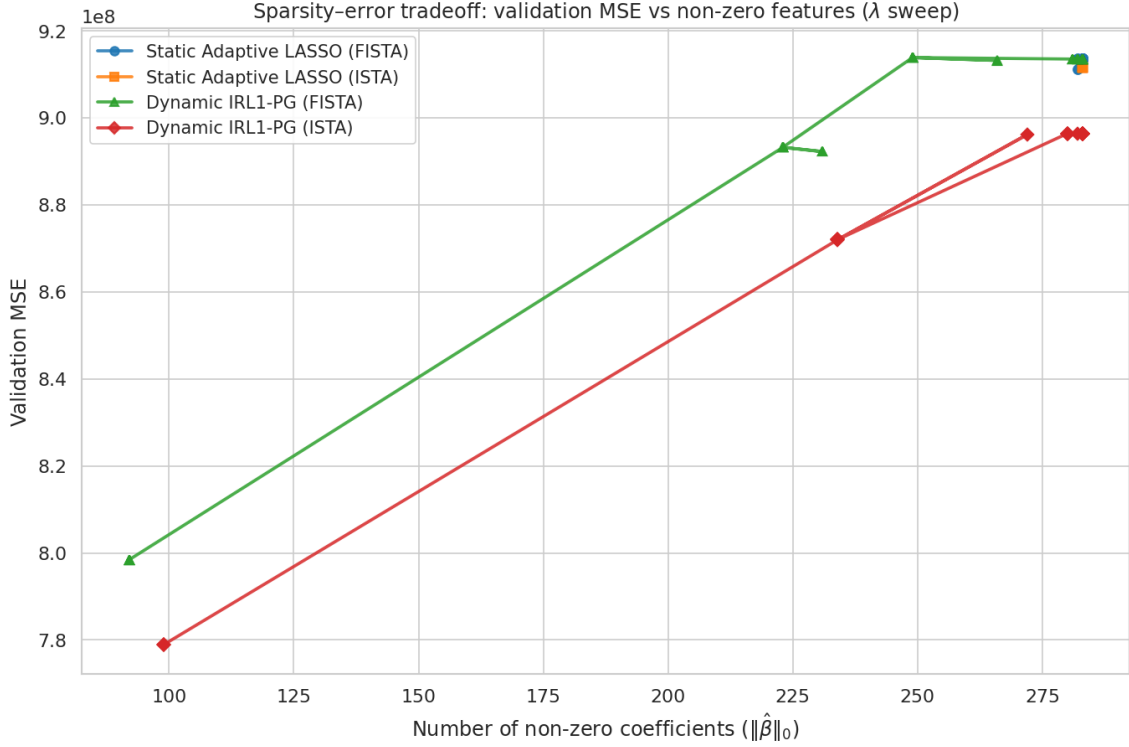


Figure 2: Validation MSE vs. number of non-zero features as λ varies. Dynamic IRL1-PG variants achieve lower error with substantially fewer active coefficients than static adaptive baselines at matched regularisation levels.

The empirical Pareto front confirms that dynamic reweighting can move to lower- ℓ_0 regions while maintaining or improving validation error versus denser static baselines.

5.4 Feature Selection Stability

Table 3: Feature selection stability loaded from `stability_table.csv`.

Method	Mean Jaccard	Std Jaccard
Adaptive LASSO	0.9873	0.0042
Dynamic IRL1-FISTA	0.2721	0.0637
Dynamic IRL1-ISTA	0.2923	0.0694
LASSO	0.7689	0.0187
Ridge	0.9958	0.0026

The feature selection stability results show a trade-off: dynamic IRL1-PG improves sparsity and test error but yields lower support overlap across seeds than static adaptive LASSO. This is consistent with pruning near correlated or borderline predictors.

5.5 Feature Interpretation

Using the dynamic FISTA run, the top absolute coefficients are dominated by interpretable housing factors from `t20_feature_interpretation.csv`. The largest positive effect is `GrLivArea`, while magnitude negative effects include `RoofMatl_ClyTile` and `Condition2_PosN`. Strong positive signals also include quality and neighborhood indicators such as `GarageQual_Ex`, `YearBuilt`, `OverallQual`, `Neighborhood_NoRidge`, and `Neighborhood_StoneBr`. This aligns with domain expectations: size, build/finish quality, and location remain the strongest determinants of sale price in the selected sparse model.

Table 4: Top 10 feature effects loaded from `t20_feature_interpretation.csv`.

Feature	Coefficient	Direction	Interpretation
<code>GrLivArea</code>	29348.34	positive	Size-related feature: larger usable space generally increases price.
<code>RoofMatl_ClyTile</code>	-21052.55	negative	General structural/categorical feature that appears to decreases price.
<code>Condition2_PosN</code>	-9516.55	negative	Quality/condition proxy: better values typically decreases price.
<code>GarageQual_Ex</code>	9370.23	positive	Quality/condition proxy: better values typically increases price.
<code>YearBuilt</code>	8778.70	positive	Age/renovation signal: newer or renovated homes usually increases price.
<code>BsmtQual_Ex</code>	8553.76	positive	Quality/condition proxy: better values typically increases price.
<code>OverallQual</code>	8356.41	positive	Quality/condition proxy: better values typically increases price.
<code>PoolArea</code>	7508.56	positive	Size-related feature: larger usable space generally increases price.
<code>GarageCond_Ex</code>	-7325.71	negative	Quality/condition proxy: better values typically decreases price.
<code>BsmtFinSF1</code>	6571.54	positive	Size-related feature: larger usable space generally increases price.

6 Limitations

1. **No global convergence theorem for the outer loop.** The ISTA/FISTA convergence rates [Beck and Teboulle, 2009] apply to the inner fixed-weight subproblems. The outer IRL1 loop changes the objective each iteration.
2. **No oracle property claim for the dynamic variant.** The static Adaptive LASSO has oracle properties under conditions specified by Zou [2006] and extended by Huang et al. [2008]. The dynamic IRL1 variant does not inherit these guarantees automatically.

3. **Efficiency claims are solver-relative.** Comparisons of runtime and iteration counts are meaningful only between custom ISTA/FISTA variants under matched stopping criteria.
4. **Sensitivity to preprocessing.** Results depend on choices in imputation, encoding, and scaling. Different preprocessing choices may alter which features are selected and how stable that selection is.
5. **Single dataset.** All empirical conclusions are specific to the House Prices dataset. Generalisation to other high-dimensional tabular regression tasks requires further evaluation.

7 Conclusion

We checked the results of IRL1-PG, the Iteratively Reweighted ℓ_1 Minimisation framework of Candès et al. [2008], combined with ISTA and FISTA proximal gradient solvers [Beck and Teboulle, 2009]. We used it for choosing important features in high-dimensional tabular regression. We applied the method to the House Prices dataset, where features are strongly correlated and dynamic reweighting is more appropriate than using a fixed ℓ_1 penalty [Zou and Hastie, 2005]. The IRL1-PG method belongs to the broader class of reweighting methods for sparsity discussed by Wipf and Nagarajan [2011], and the FISTA speedup follows Nesterov’s momentum idea [Nesterov, 1983].

The main findings are as follows:

- IRL1-PG gives sparser models, using only 92–99 features, compared with Ridge, LASSO, and static Adaptive LASSO, which use 240–283 features. IRL1-PG also performs better on test error for this dataset.
- Within IRL1-PG variants, ISTA performs best on validation MSE, while FISTA performs best on test MSE and test MAE, and both have similar runtime.
- Dynamic reweighting makes models sparser, but also makes support selection less stable across runs with different random seeds compared with fixed weights.

We also note practical limits:

- We do not claim global convergence of the outer IRL1-PG loop, and results can still be affected by preprocessing choices.

In future work, we plan to extend IRL1-PG to more general model classes, combine it with group-sparse penalties for structured feature correlations [Zou and Hastie, 2005], and apply it to larger genomics or finance datasets where the number of features is much larger than the number of samples [Huang et al., 2008].

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